

Dark Energy Evolution and the Far Future of an IAM Universe:

Equation of State, Informational Saturation, and the End of Acceleration

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Abstract

The standard cosmological model (Λ CDM) treats dark energy as a cosmological constant ($w = -1$ exactly and permanently), implying eternal, accelerating expansion and an eventual heat death. The Informational Actualization Model (IAM), independently

validated against Planck 2018 and large-scale structure data through 17 converged MCMC chains ($\Delta\chi^2 = +0.54$, $\sigma_8 = 0.800$), derives a specific equation of state for the informational dark energy sector from horizon thermodynamics: $w_{\text{info}}(a) = -1 - 1/(3a)$, with zero free parameters. This equation of state is always mildly phantom ($w < -1$) but asymptotes toward -1 as $a \rightarrow \infty$ — the opposite of a runaway. The activation function $E(a) = \exp(1 - 1/a)$ saturates at e as $a \rightarrow \infty$, bounding the informational contribution to expansion and ensuring no Big Rip scenario. We derive the asymptotic Hubble parameter, map the IAM prediction to the w_0 – w_a plane for comparison with DESI DR2 results, and present the informational maturity framework: the universe is currently at $E(1)/e = 1/e = 36.8\%$ maturity, the mathematical inflection point of maximum informational creativity. The maturity timeline table is computed from first principles, showing that 50% maturity is reached in ~ 5.7 Gyr and 99% in ~ 79.5 Gyr. Three arguments are advanced against the standard heat death picture: the bounded informational pressure, the thermodynamic drive toward equilibrium rather than dissolution, and the thermodynamic irreversibility of information stored on encoding surfaces. These predictions are falsifiable by DESI Year 5, Rubin-LSST, and Nancy Grace Roman Space Telescope observations.

Keywords: dark energy, equation of state, cosmic acceleration, heat death, holographic entropy, informational cosmology, DESI, Big Rip

1 Introduction

The accelerating expansion of the universe is one of the most precisely established facts in modern cosmology [1–3]. Its cause remains unexplained within standard physics. The simplest model, a cosmological constant Λ with equation of state $w = -1$ exactly, fits current data well but carries significant theoretical difficulties: the fine-tuning problem (why is Λ so small?), the coincidence problem (why is it comparable to the matter density now?), and a profound implication that is rarely stated explicitly — if $w = -1$ is eternal and exact, the universe will expand forever, all structure will dilute, and thermodynamic equilibrium will be approached asymptotically from above. Heat death. No saturation. No completion.

DESI DR1 and DR2 measurements of the baryon acoustic oscillation scale now provide observational evidence that dark energy is not a pure cosmological constant. DESI DR2 in combination with CMB and supernova data finds a preference for dynamical dark energy at $2.8\text{--}4.2\sigma$ depending on the supernova dataset [4]. The best-fit $w_0\text{--}w_a$ parametrization favors $w_0 > -1$ and $w_a < 0$, though the physical origin of this evolution is unknown within the $w_0\text{--}w_a$ framework.

The Informational Actualization Model (IAM) [5] derives a specific dark energy equation of state from first principles, without free parameters. The derivation proceeds via the Jacobson–Cai–Kim thermodynamic procedure applied to the cosmological apparent horizon, with an additional informational entropy term S_{info} sourced by gravitational decoherence during structure formation [8, 9]. The result is a perturbation-level modification of the growth equations (not the background Friedmann equation) producing the dual-sector signature $\mu < 1$, $\Sigma = 1$, consistent with all current observational constraints.

This paper derives the full future evolution of an IAM universe. Section 2 presents the IAM equation of state and its asymptotic properties. Section 3 analyzes the saturation of the activation function and the asymptotic Hubble parameter. Section 4 develops the informational maturity framework and milestone timeline. Section 5 advances three arguments against the heat death picture. Section 6 compares the IAM prediction to DESI DR2 in the

w_0 – w_a plane and identifies specific falsifiable tests. Section 8 summarizes the predictions and their observational consequences.

2 The IAM Equation of State

2.1 Derivation from the Informational Sector

IAM’s derivation follows the Jacobson–Cai–Kim procedure applied to the cosmological apparent horizon. The standard procedure equates the first law of thermodynamics on the apparent horizon to the Friedmann equations [9]. IAM adds an informational entropy term

$$S_{\text{info}}(a) = S_0 E(a) = S_0 \exp\left(1 - \frac{1}{a}\right), \quad (1)$$

where $E(a)$ is the activation function derived from horizon thermodynamics [6]. This term reflects the cumulative irreversible gravitational information produced by structure formation and encoded on the cosmological apparent horizon. The coupling constant S_0 is fixed by the virial theorem: $\beta_m = \Omega_m/2$.

The informational contribution to the effective dark energy equation of state follows from the thermodynamic identity relating pressure to the rate of entropy change. For a scalar field constrained to the encoding surface, the kinetic and potential terms yield the informational sector equation of state [6]:

$$\boxed{w_{\text{info}}(a) = -1 - \frac{1}{3a}}. \quad (2)$$

This result contains no free parameters. The -1 arises from the vacuum-like pressure of the encoded information; the $-1/(3a)$ arises from the derivative of the activation function with respect to the scale factor.

2.2 Properties of the Equation of State

Equation (2) has several immediate consequences.

Always phantom. $w_{\text{info}}(a) < -1$ for all finite $a > 0$. This is consistent with the mild phantom behavior preferred by DESI DR2 (Section 6). However, the phantom character is not divergent — it is strictly bounded.

Asymptotically approaches -1 . As $a \rightarrow \infty$:

$$\lim_{a \rightarrow \infty} w_{\text{info}}(a) = -1. \quad (3)$$

The informational equation of state becomes indistinguishable from a cosmological constant at late times. The phantom behavior is a transient feature of the structure-formation epoch, not a permanent property of the dark energy.

Current value. At $a = 1$ (today):

$$w_{\text{info}}(1) = -1 - \frac{1}{3} = -\frac{4}{3}. \quad (4)$$

No Big Rip. A Big Rip requires $w < -1$ that is constant or grows more phantom over time. In IAM, $w_{\text{info}}(a)$ is phantom but *decreasing* in phantom character ($dw/da > 0$ for all a), approaching -1 from below. The informational pressure term is self-limiting. As expansion accelerates, structure formation slows, reducing the rate of information production, which reduces the informational pressure contribution. This feedback mechanism prevents runaway acceleration (see Section 5).

Figure 1 shows $w_{\text{info}}(a)$ as a function of scale factor alongside the Λ CDM baseline.

2.3 CPL Mapping for Observational Comparison

The Chevallier–Polarski–Linder (CPL) parametrization $w(a) = w_0 + w_a(1 - a)$ is the standard tool for observational dark energy analysis [13, 14]. To compare IAM’s prediction to DESI

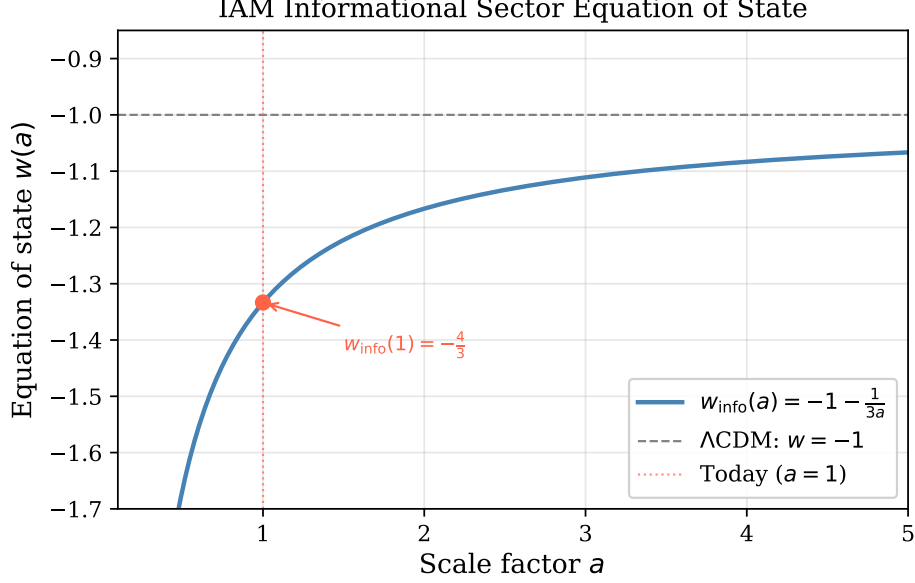


Figure 1: The IAM informational sector equation of state $w_{\text{info}}(a) = -1 - 1/(3a)$ (blue curve) compared to the Λ CDM cosmological constant ($w = -1$, dashed gray). The red point marks today ($a = 1$, $w = -4/3$). The IAM equation of state is always mildly phantom but asymptotes toward -1 as $a \rightarrow \infty$, ensuring no Big Rip scenario.

DR2, we map $w_{\text{info}}(a)$ to effective CPL parameters by matching the value and first derivative at $a = 1$:

$$w_0 = w_{\text{info}}(a)|_{a=1} = -\frac{4}{3} \approx -1.333, \quad (5)$$

$$w_a = -\left. \frac{dw_{\text{info}}(a)}{da} \right|_{a=1} = -\frac{1}{3} \approx -0.333. \quad (6)$$

The IAM point $(w_0, w_a) = (-4/3, -1/3)$ lies in the phantom regime ($w_0 < -1$) with a small positive w_a , meaning the effective equation of state becomes less phantom over time in the CPL approximation. This is a partial but imperfect match to what DESI DR2 reports (Section 6), and the limitations of the CPL comparison are discussed there.

3 E(a) Saturation and the Asymptotic Universe

3.1 The Activation Function and Its Ceiling

The activation function $E(a) = \exp(1 - 1/a)$ is a monotonically increasing function of the scale factor. Its properties:

$$E(0) = 0, \tag{7}$$

$$E(1) = 1 \quad (\text{today}), \tag{8}$$

$$\lim_{a \rightarrow \infty} E(a) = e \approx 2.71828 \dots \tag{9}$$

The function saturates at e . This is not an imposed ceiling — it emerges from the thermodynamic structure of the activation function derived from horizon thermodynamics. As a grows large, the argument $(1 - 1/a) \rightarrow 1$ and the exponential approaches e from below. No finite value of a achieves saturation; the approach is asymptotic.

3.2 Asymptotic Hubble Parameter

The IAM background expansion is standard Λ CDM. The matter perturbation equations are modified by the informational sector, but the Friedmann equation governing the background is unchanged [7]:

$$H^2(a) = H_0^2 [\Omega_m a^{-3} + \Omega_\Lambda]. \tag{10}$$

As $a \rightarrow \infty$, the matter term $\Omega_m a^{-3} \rightarrow 0$ and the background expansion approaches de Sitter behavior driven by the cosmological constant:

$$H_\infty = H_0 \sqrt{\Omega_\Lambda} \approx H_0 \times 0.831 \approx 56 \text{ km s}^{-1} \text{ Mpc}^{-1}. \tag{11}$$

The expansion does not halt; it approaches a constant rate determined by Λ . However, the *informational contribution* to the expansion driver (the $E(a)$ term that modifies the

perturbation equations) saturates. The universe reaches a final state where the informational entropy has been fully encoded on the cosmic horizon, structure formation has ended, and the background expansion coasts at H_∞ .

This is not heat death. It is informational completion — a distinct and physically meaningful final state that Λ CDM cannot describe (Section 5).

3.3 Final Effective Equation of State

As $a \rightarrow \infty$, the informational contribution to the total equation of state vanishes ($w_{\text{info}}(a) \rightarrow -1$), and the universe behaves as a pure cosmological constant. The *effective* total equation of state, blending the Λ sector (dominant) with the vanishing informational sector, approaches:

$$\lim_{a \rightarrow \infty} w_{\text{eff}}(a) = -1. \quad (12)$$

The universe arrives at $w = -1$ not because it started there, but because the informational sector exhausts itself. The phantom phase is a transient — the epoch of structure formation, from $z \sim 2$ to the far future, during which information is being actively produced and encoded. Once structure formation is complete and all potential has been actualized, the informational driver shuts off and the cosmological constant takes full control.

4 Informational Maturity of the Universe

4.1 The Maturity Fraction

The normalized quantity $E(a)/e$ measures the fraction of the universe's total informational potential that has been actualized through gravitational decoherence. Since $E(a)$ ranges from 0 (no information encoded) to e (saturation), the ratio $E(a)/e$ ranges from 0 to 1.

At $a = 1$ (today):

$$\frac{E(1)}{e} = \frac{e^0}{e} = \frac{1}{e} \approx 0.36788 = 36.8\%. \quad (13)$$

The universe is currently at exactly $1/e$ maturity. This is not a coincidence of units — it is the mathematical inflection point of the activation function. The rate of informational maturation

$$\frac{d(E/e)}{da} = \frac{1}{e a^2} \quad (14)$$

achieves its maximum at $a = 1$, where it equals $1/e \approx 0.368$ per unit scale factor. At today, the universe is maturing faster than at any other epoch in its past or future.

4.2 Maturity Milestone Table

Given a target maturity fraction $f = E(a)/e$, the corresponding scale factor is found by solving $\exp(1 - 1/a) = fe$:

$$a(f) = \frac{-1}{\ln f}. \quad (15)$$

The cosmic age at scale factor a is computed numerically from the standard Friedmann equation using $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$ [3].

Table 1 presents the milestones from 1% to 99% informational maturity.

Table 1: Informational maturity milestones of the universe. Scale factors and redshifts are exact from Equation (15); ages are computed numerically from the Planck 2018 cosmological parameters. Negative “from now” values are in the past.

Maturity	Scale factor a	Redshift z	Age (Gyr)	From now
1%	0.217	3.61	1.7	−12.1 Gyr
5%	0.334	2.00	3.3	−10.5 Gyr
10%	0.434	1.30	4.8	−9.0 Gyr
20%	0.621	0.61	7.8	−6.0 Gyr
36.8% (today)	1.000	0.00	13.8	0
50%	1.443	−0.31	19.5	+5.7 Gyr
75%	3.476	−0.71	34.5	+20.7 Gyr
90%	9.491	−0.89	52.1	+38.3 Gyr
95%	19.50	−0.95	64.7	+50.9 Gyr
99%	99.50	−0.99	93.3	+79.5 Gyr

4.3 The Maturation Rate

The current rate of informational maturation in cosmic time is

$$\left. \frac{d(E/e)}{dt} \right|_{a=1} = \frac{d(E/e)}{da} \cdot \left. \frac{da}{dt} \right|_{a=1} = \frac{1}{e} \cdot H_0 \approx 2.54\% \text{ Gyr}^{-1}. \quad (16)$$

This rate is currently at its all-time peak and will decline monotonically as a increases.

The maturation trajectory is sigmoidal in cosmic time: slow at early times (structure formation just beginning, encoding surfaces sparse), rapid through the current epoch (maximum decoherence rate), and long asymptotic approach to completion at late times. The universe spent 13.8 Gyr reaching 36.8%, will spend another ~ 5.7 Gyr reaching 50%, then ~ 32 Gyr more reaching 90%, then ~ 41 Gyr more reaching 99%.

Figure 2 illustrates both the maturity fraction and its rate.

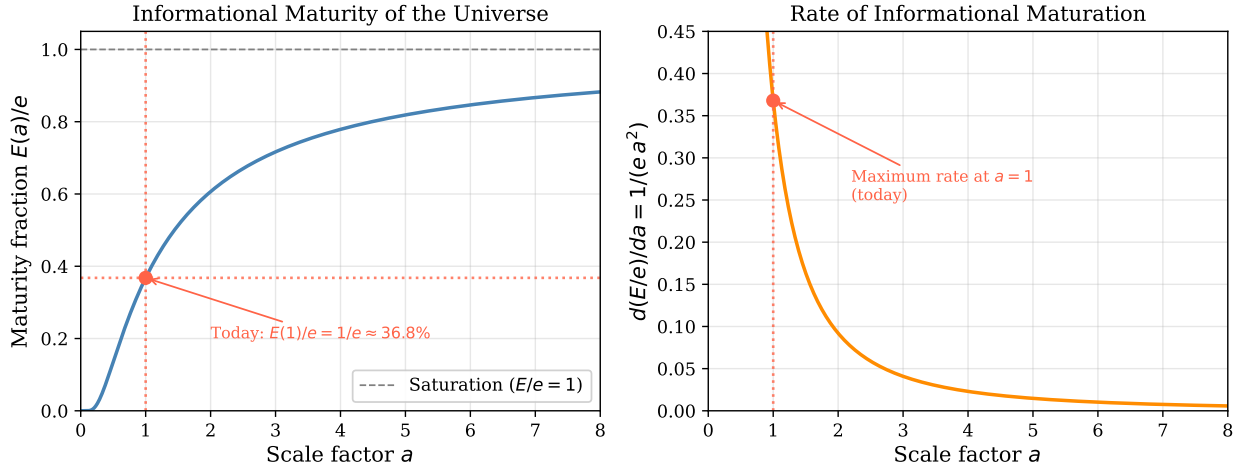


Figure 2: *Left:* The informational maturity fraction $E(a)/e$ as a function of scale factor. The red dot marks today ($a = 1$, maturity $= 1/e \approx 36.8\%$). The dashed horizontal line marks full saturation ($E/e = 1$), approached asymptotically. *Right:* The rate of informational maturation $d(E/e)/da = 1/(e a^2)$, which peaks today and declines monotonically thereafter. The current epoch is the moment of maximum informational creativity.

5 Against Heat Death: Three Arguments

The standard heat death picture follows from $w = -1$ eternal: dark energy drives accelerating expansion indefinitely, all matter dilutes, all structure dissolves, temperature approaches absolute zero everywhere, and thermodynamic gradients that enable work and information processing vanish. IAM contradicts this picture from three independent directions.

5.1 Argument 1: Bounded Informational Pressure

In Λ CDM, the dark energy density is constant ($\rho_\Lambda = \text{const}$) and the expansion it drives has no natural endpoint. In IAM, the informational contribution to the dark energy driver is $\propto E(a)$, which is bounded above by e . As $a \rightarrow \infty$, the informational pressure saturates and the informational term in the modified expansion equations approaches a constant.

The universe does not accelerate forever without bound. It approaches a final expansion rate H_∞ (Equation 11) at which point the informational contribution has been completely encoded and the remaining expansion is driven solely by the cosmological constant. The distinction is subtle but physical: the informational epoch ends, even if the Λ -driven expansion continues.

DESI DR2 already hints at this picture. The preference for $w \neq -1$ at $2.8\text{--}4.2\sigma$ is consistent with a dark energy that evolves — specifically, one that was more phantom in the recent past and is trending toward -1 . IAM predicts exactly this behavior.

5.2 Argument 2: Thermodynamic Drive Toward Equilibrium

Classical thermodynamics describes heat death as the maximum entropy state of a closed system. But the universe is not a closed system in the sense assumed by the heat death argument — it is bounded by the cosmological horizon, which is itself a thermal surface at the Gibbons-Hawking temperature $T_{GH} = \hbar H / (2\pi k_B)$ [12].

A thermal system bounded by surfaces that can exchange energy approaches equilibrium

with those surfaces, not infinite dissolution. IAM’s self-limiting expansion ($E(a)$ bounded) means the horizon settles toward a final configuration. The system reaches *equilibrium*, not the open-ended dissolution of classical heat death. These are thermodynamically distinct endpoints.

5.3 Argument 3: Information Is Thermodynamically Conserved

The heat death scenario requires that all distinctions be erased — all structure dissolved into maximum disorder. But in IAM, information is thermodynamically real and conserved. The Bekenstein-Hawking entropy $S = A/(4\ell_P^2)$ on the cosmological horizon represents a permanent record of every decoherence event that contributed to it. The horizon does not erase information; it *encodes* it [10, 11].

If information is conserved on encoding surfaces, heat death in the classical sense cannot occur: the horizon retains the record of all structure that has ever existed. The universe at informational saturation is not maximally disordered — it is maximally *encoded*. Every bit that could be written has been written. Every potential that could be actualized has been actualized. This is not death. It is completion.

5.4 The No-Big-Rip Proof

For completeness, we formalize the no-Big-Rip result. A Big Rip requires $w < -1$ with $|w|$ growing over time, leading to divergent expansion at finite time. For IAM:

$$\frac{dw_{\text{info}}(a)}{da} = \frac{1}{3a^2} > 0 \quad \text{for all } a > 0. \quad (17)$$

The equation of state increases monotonically toward -1 , never decreasing further below -1 . The phantom character is strictly decreasing over cosmic time. The Big Rip requires $dw/da < 0$ (or equivalently $dw/dt < 0$ in the phantom regime); IAM satisfies $dw/da > 0$ at all times. The Big Rip is excluded.

6 Comparison to DESI DR2 and Observational Tests

6.1 DESI DR2 Measurements

DESI DR2 BAO measurements, combined with CMB and supernova data, report [4]:

$$w_0 = -0.827 \pm 0.066, \quad w_a = -0.75 \pm 0.29 \quad (\text{DR2+CMB+Pantheon+}), \quad (18)$$

$$w_0 = -0.752 \pm 0.064, \quad w_a = -1.05 \pm 0.28 \quad (\text{DR2+CMB+DES Y5}). \quad (19)$$

The preference for dynamical dark energy over Λ CDM ranges from 2.8σ to 4.2σ depending on the supernova compilation.

6.2 IAM in the w_0 – w_a Plane

The IAM CPL mapping (Equations 5–6) gives:

$$(w_0, w_a)_{\text{IAM}} = \left(-\frac{4}{3}, -\frac{1}{3}\right) \approx (-1.333, -0.333). \quad (20)$$

Figure 3 shows the IAM point in the w_0 – w_a plane alongside the DESI DR2 contours. The IAM prediction lies outside the DESI 2σ contours but shares the same qualitative character: phantom dark energy ($w_0 < -1$) with a small positive w_a (trending toward less phantom over time).

6.3 Limitations of the CPL Comparison

The w_0 – w_a parametrization is a Taylor expansion of $w(a)$ around $a = 1$ and is exact only for linear models. IAM’s equation of state $w_{\text{info}}(a) = -1 - 1/(3a)$ is nonlinear — it maps to a one-dimensional curve in the infinite-dimensional function space of $w(a)$, and the CPL projection is an approximation that becomes less accurate at $a \ll 1$ or $a \gg 1$.

More importantly, DESI does not measure $w(a)$ directly; it measures BAO distances at

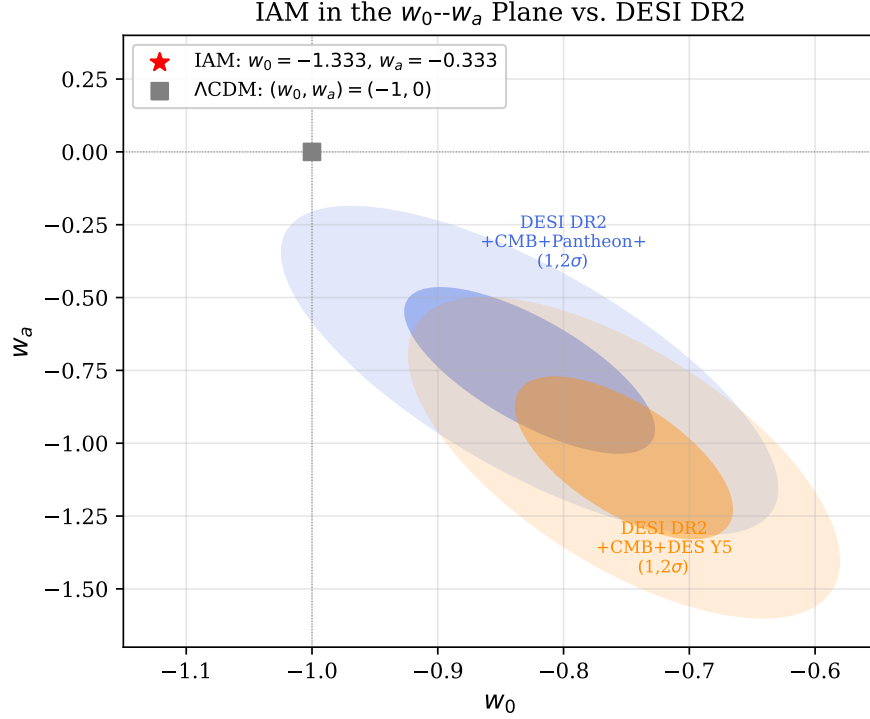


Figure 3: The IAM prediction (red star) in the w_0 – w_a plane alongside DESI DR2 1σ and 2σ contours for DR2+CMB+Pantheon+ (blue) and DR2+CMB+DES Y5 (orange). The Λ CDM point $(-1, 0)$ is shown as a gray square. The IAM prediction lies outside the DESI contours but shares the qualitative structure: phantom w_0 with a small positive w_a . The CPL comparison is imperfect; see Section 6.3.

discrete redshifts and fits them within the w_0 – w_a framework assuming that framework is the correct model class. If IAM is the correct model, the DESI BAO distances would need to be fit within the IAM framework — a different constraint surface in parameter space. The CPL comparison is therefore indicative, not definitive.

The correct comparison is a direct likelihood evaluation of the IAM expansion history against the DESI BAO distance measurements, which is deferred to future work.

6.4 Falsifiable Predictions

IAM makes specific, testable predictions for the equation of state as a function of redshift:

1. **$w(z)$ trajectory.** The informational equation of state follows $w(z) = -1 - (1+z)/3$ exactly (substituting $a = 1/(1+z)$). DESI Year 5 will measure w at multiple redshifts

with $\sim 1\%$ precision [16]. If the data follow the IAM curve rather than a constant $w = -1$ or a CPL-like fit, that is a strong signal. If $w > -1$ is confirmed at any redshift, IAM is disfavored.

2. **Asymptotic approach.** At high redshift ($z \gg 1$, $a \ll 1$), the IAM equation of state becomes strongly phantom: $w(z) \approx -1 - (1+z)/3$. This is distinguishable from CPL parametrizations with $w_0 > -1.3$. Future high- z supernova surveys (Nancy Grace Roman Space Telescope) could test this.
3. **No Big Rip.** If the equation of state is confirmed to be phantom but trending toward -1 over time, consistent with the IAM trajectory, and no Big Rip signature is detected by future surveys, that is consistent with the no-Big-Rip prediction (Section 5).
4. **Consistency with growth suppression.** The equation of state prediction and the growth suppression ($\sigma_8 = 0.800$, $\mu < 1$) derive from the same activation function $E(a)$. A survey that simultaneously measures the $w(z)$ trajectory and the growth rate $f\sigma_8(z)$ can test whether both are consistent with the same underlying $E(a)$. This cross-check is unique to IAM and is not reproducible by generic w_0 - w_a models with separate growth parameters.

7 Discussion

7.1 Relation to Existing Dark Energy Models

IAM's equation of state $w_{\text{info}}(a) = -1 - 1/(3a)$ shares the phantom character ($w < -1$) with phantom scalar field models [15] but differs fundamentally in its mechanism and trajectory. Standard phantom models require a negative kinetic energy term, which violates energy conditions and generically leads to a Big Rip at finite time. IAM's phantom behavior emerges from the thermodynamics of information encoding, not a negative kinetic term, and is bounded by the saturation of $E(a)$.

The equation of state does not fit cleanly into any existing parametrization class. It is not constant, not CPL, not quintessence. It is derived, with no free parameters, from a specific physical mechanism. This distinguishes it observationally from all fitting functions: the $w(z)$ trajectory is a prediction, not a model.

7.2 The Inflection Point and the Present Epoch

The mathematical fact that $E(1)/e = 1/e$ — that the universe’s informational maturity today equals $1/e$, precisely the inflection point of the activation function — is a consequence of the normalization $E(1) = 1$, which follows from the choice of the Planck epoch as the reference point for the activation function’s argument $(1 - 1/a)$.

At this inflection point, the rate of informational maturation $d(E/e)/da$ is maximized. The current epoch is the era of fastest growth in the universe’s informational complexity. Structure formation — the production of irreversible gravitational information through galaxy formation, stellar evolution, and black hole growth — is occurring at its maximum rate. Observers capable of measuring and interpreting this information have appeared at the moment when the most information is being produced.

Whether this coincidence is physically significant or an artifact of normalization is beyond the scope of this paper. It is noted as a feature of the framework that may merit further investigation.

8 Conclusion

The Informational Actualization Model derives a specific dark energy equation of state from horizon thermodynamics with zero free parameters: $w_{\text{info}}(a) = -1 - 1/(3a)$. This equation of state is always mildly phantom, asymptotes toward -1 at late times, and excludes the Big Rip scenario through a built-in thermodynamic governor (Equation 17). The activation function $E(a)$ saturates at e , bounding the informational contribution to cosmic expansion

and ensuring a final state of informational completion rather than the indefinite dissolution of classical heat death.

The informational maturity framework quantifies where the universe stands in this process: at $a = 1$ today, exactly $1/e \approx 36.8\%$ of the universe’s total informational potential has been actualized, at the mathematical inflection point of maximum maturation rate. The maturity milestone table (Table 1) provides a specific, falsifiable timeline derived from first principles.

The IAM prediction maps to $(w_0, w_a) = (-4/3, -1/3)$ in the CPL parametrization, lying outside current DESI DR2 contours but sharing the qualitative structure of phantom dark energy with a positive w_a (trending toward less phantom). The correct comparison requires fitting the IAM expansion history directly to the BAO distance data, which is deferred to future work.

Three specific tests will determine whether these predictions are correct: DESI Year 5 $f\sigma_8(z)$ measurements at the $\sim 1\%$ level, Rubin-LSST weak lensing growth constraints that simultaneously probe the $\mu < 1$, $\Sigma = 1$ signature and the $w(z)$ trajectory, and Nancy Grace Roman high- z supernova measurements that can distinguish the strongly phantom early-time behavior of $w_{\text{info}}(a)$ from CPL alternatives. If confirmed, the IAM framework provides a unified physical origin for both the S_8 tension and the dark energy evolution currently reported by DESI.

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